Study-aware QSPR of human skin permeability: separating molecular signal from source heterogeneity and calibrating unseen sources

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1 Abstract

Human skin permeability measurements are commonly pooled across studies even though permeability depends not only on molecular properties but also on study-specific experimental conditions. We tested whether this heterogeneity can be modeled as a reusable source-associated component rather than treated solely as noise. Three public human-skin permeability resources were used sequentially rather than pooled: the Cheruvu et al. 2022 dataset for development, HuSkinDB for independent replication and held-out-source calibration, and SkinPiX for cross-dataset transfer. Measurement-level mixed-effects models separated molecular, residual compound-specific, literature-source, and residual variation. The molecular component was deliberately simple and prespecified, using molecular weight and logP alone (S1) or a six-descriptor physicochemical specification (S2). In Cheruvu et al., source-associated variation accounted for 22.8% (95% CI 14.4–31.9%) of modeled random/residual variance; in HuSkinDB it accounted for 44.4% (33.5–54.5%). Source-blind hierarchical prediction did not show a reproducible improvement over conventional aggregation: RMSE improvement was 5.4% (95% CI -3.6–10.9%) for Cheruvu S1, 7.0% (-5.1–14.0%) for Cheruvu S2, and -1.7% (-5.1–1.0%) in HuSkinDB. In contrast, known-source prediction improved RMSE by 16.7% (11.6–21.8%), 18.7% (11.5–23.3%), and 17.0% (10.0–23.8%), respectively. When entire HuSkin sources were held out, offsets estimated from shared anchor compounds improved prediction of compounds absent from model fitting; among 9 matched sources, mean source-level RMSE reduction increased from 31.1% (13.4–42.1%) with 1 anchor to 45.3% (19.2–60.0%) with 5 anchors. A falsification control showed that source-matched calibration consistently outperformed offsets borrowed from another source. Finally, one-anchor calibration reduced mean source-level RMSE by 17.6% across 4 independent SkinPiX sources containing 54 HuSkin-unseen compounds, all within the HuSkin-defined chemical applicability domain. These findings support a study-aware QSPR framework in which literature-source heterogeneity is modeled explicitly and can be estimated for a new experimental source using a small number of shared compounds.

Keywords: skin permeability; QSPR; mixed-effects model; hierarchical model; source calibration; external transfer; applicability domain; HuSkinDB; SkinPiX

2 Introduction

The skin permeability coefficient, K_p, is a central parameter in transdermal drug delivery, dermal toxicology, cosmetics, and exposure assessment. Because direct measurement in human skin is expensive, tissue-limited, and experimentally demanding, quantitative structure–property relationship (QSPR) models have long been used to estimate permeability from molecular structure. The classical Potts–Guy model relates permeability primarily to lipophilicity and molecular size (Potts and Guy 1992), while later studies have explored broader physicochemical descriptor sets and more flexible statistical or machine-learning methods (Tsakovska et al. 2017; Zeng et al. 2021).

A major complication is that the published literature does not represent a single experimental system. Human-skin permeability values are generated across laboratories, diffusion-cell configurations, skin sources and anatomical sites, tissue-preparation procedures, donor and receptor media, temperatures, analytical methods, and reporting conventions. Interstudy variation has therefore been recognized for decades. Fujiwara et al. modeled a latent common permeability underlying study-specific measurements (Fujiwara et al. 2003). Abraham and Martins standardized literature values for temperature and ionization before fitting linear free-energy relationships (Abraham and Martins 2004). Samaras et al. showed that experimental and formulation conditions materially influence in-vitro human skin permeation (Samaras et al. 2012). More recently, HuSkinDB and SkinPiX have made measurement-level permeability records and protocol metadata available in structured public resources (Stepanov et al. 2020; Chedik et al. 2024).

Recent predictive studies further narrow the remaining methodological gap. QSPR models have been developed from curated HuSkinDB records (Waters and Quah 2022), and newer work has combined HuSkinDB and SkinPiX into larger curated resources for predictive modeling (Asgarkhanova et al. 2026). Oh and Ahn explicitly examined disagreement among repeated permeability measurements, protocol variables, record-level versus compound-level modeling, and mixed-effects formulations (Oh and Ahn 2026). Consequently, novelty can no longer rest on the general observations that repeated measurements disagree, that experimental conditions matter, or that mixed-effects models can be fitted to permeability data.

The more operational question is whether source-associated heterogeneity can be **estimated for a previously unseen experimental source and then used to improve predictions for new compounds measured by that source**. If measurements can be separated into a transferable molecular component and a source-associated component shared across compounds, then compounds measured both in a reference corpus and on a new platform could serve as calibration anchors. Under this interpretation, study heterogeneity becomes not only a source of uncertainty but also a parameter that can be estimated when deploying a model to a new laboratory, apparatus, or protocol.

We therefore considered the measurement-level model

$$y_{ij} = f(X_i) + \beta^\top Z_{ij} + b_i + u_j + \epsilon_{ij},$$

where y_{ij} is log permeability for compound i measured in source j, $f(X_i)$ is the systematic molecular component, Z_{ij} contains measured experimental covariates, b_i is residual compound-specific variation not explained by $f(X_i)$, u_j is a source-associated random intercept, and ϵ_{ij} is measurement-level residual error. For a previously measured compound, the total chemical contribution is $f(X_i) + b_i$; for a genuinely unseen compound, only $f(X_i)$ is available in expectation.

The analysis was designed as development, independent replication, and cross-dataset transfer. The Cheruvu et al. dataset (Cheruvu et al. 2022) was used to establish the variance structure and compare conventional compound aggregation with measurement-level hierarchical prediction. HuSkinDB was then used to replicate the hierarchy and to simulate deployment by holding out entire literature sources. Finally, SkinPiX was kept separate as an external transfer set: the molecular model was trained on HuSkinDB only, shared HuSkin–SkinPiX compounds were used as anchors, and SkinPiX-only compounds were reserved for evaluation. The central hypotheses were that (i) permeability measurements contain a reproducible source-associated component not captured by molecular descriptors alone, and (ii) this component can be estimated from a small number of shared compounds to improve prediction at an unseen source.

3 Materials and Methods

3.1 Study design and data sources

Three public human-skin permeability datasets were analyzed sequentially (Table 1). The Cheruvu et al. 2022 compilation was used for model development (Cheruvu et al. 2022) and contained 476 retained measurements from 148 compounds and 69 literature references. HuSkinDB was used for independent replication and internal

source-calibration experiments (Stepanov et al. 2020); the processed dataset contained 546 measurements from 251 compounds and 95 references. SkinPiX was used only for cross-dataset transfer (Chedik et al. 2024), with 199 retained measurements from 108 compounds and 36 references after preprocessing. Dataset-specific exclusions and provenance audits are reported in the supplementary material.

Table 1: Dataset structure and role in the analysis.

Dataset	Measurements	Compounds	Sources	Repeated compounds	Multi-source compounds	Role
Cheruvu	476	148	69	81	46	Development
HuSkinDB	546	251	95	89	61	Independent replication and held-out-source calibration
SkinPiX	199	108	36	65	14	Cross-dataset transfer validation

The datasets were deliberately **not pooled before validation**. Their separation allowed the analysis to distinguish development from replication and to evaluate transfer to an independently curated database. This design also reduced the risk that apparently external performance would be driven by duplicate literature sources present in more than one database.

3.2 Harmonization and compound/source identities

In the Cheruvu dataset, canonical compound identity was based primarily on normalized compound name rather than CAS number because the source table contained at least one CAS collision between chemically distinct entries. Reference identity was taken from the literature-source field. Permeability values were analyzed on the supplied log scale. Experimental temperature values reported in Kelvin were converted to degrees Celsius.

HuSkinDB compound identity was defined using SMILES where available, with normalized compound names used as fallback. Literature sources were identified preferentially by DOI and otherwise by the database reference field. HuSkinDB reports log K_p in cm/s; values were converted to $\log_{10}$(cm/h) by adding $\log_{10}(3600)$ so that the endpoint was compatible with the other analyses.

For SkinPiX, compound identity was defined from SMILES with normalized names as fallback. Source identity was based on publication identifier and DOI. Exact equality measurements were retained; censored inequality records were excluded rather than treated as exact K_p values. Cross-database HuSkin–SkinPiX compound overlap was identified first by exact SMILES and then by unambiguous normalized-name matching. All resulting overlap assignments were manually audited. No DOI overlap was identified between the HuSkin and SkinPiX sources used for transfer.

Because HuSkinDB and SkinPiX did not provide identical molecular-descriptor definitions, molecular weight and Crippen logP were recomputed from SMILES using RDKit for **both** databases (Landrum et al. 2026). All HuSkin and SkinPiX structures parsed successfully. Descriptor concordance among matched compounds was verified in the transfer audit, supporting compatibility of the recomputed molecular descriptors across databases.

3.3 Hierarchical decomposition in the development dataset

The initial Cheruvu hierarchy was examined using nested linear mixed-effects models fitted with `lme4` (Bates et al. 2015):

$$
\begin{aligned}
M_0 &: y_{ij} = \alpha + b_i + \epsilon_{ij}, \\
M_1 &: y_{ij} = \alpha + b_i + u_j + \epsilon_{ij}, \\
M_2 &: y_{ij} = \alpha + \beta_T T_{ij} + b_i + u_j + \epsilon_{ij}, \\
M_3 &: y_{ij} = \alpha + \beta_T T_{ij} + \beta_H H_{ij} + b_i + u_j + \epsilon_{ij},
\end{aligned}
$$

where T_{ij} is donor/skin-surface temperature and H_{ij} is skin thickness. Compound and reference were modeled as crossed random intercepts. Maximum-likelihood fits were used to compare fixed-effect specifications; restricted maximum likelihood (REML) fits were used for final parameter and variance-component estimation. For presentation, the compound, source, and residual variance components were expressed as proportions of their sum, $\sigma_b^2 + \sigma_u^2 + \sigma_\epsilon^2$. These percentages therefore describe the partitioning of modeled random/residual variance and do not include variation attributable to fixed effects. Because testing a variance component against zero places the null hypothesis on the parameter boundary, the addition of the reference random effect was interpreted primarily using the large change in information criteria and the estimated variance component rather than relying on a naive chi-square test. Temperature and skin-thickness additions were compared using ordinary likelihood-ratio tests.

3.4 Molecular QSPR specifications

The molecular component $f(X)$ was prespecified rather than selected by stepwise search. Two specifications were evaluated in Cheruvu:

S1: size–lipophilicity model

$$
f_{S1}(X_i) = \beta_0 + \beta_1 MW_i + \beta_2 \log K_{ow,i}.
$$

S2: extended physicochemical model

$$
f_{S2}(X_i) = \beta_0 + \beta_1 MW_i + \beta_2 \log K_{ow,i} + \beta_3 MP_i + \beta_4 \log S_{aq,i} + \beta_5 HBD_i + \beta_6 HBA_i.
$$

Descriptors were standardized using compound-level training-set means and standard deviations. The molecular terms were fitted jointly with the selected hierarchical structure rather than estimated from pre-aggregated permeability values. The explanatory contribution of $f(X)$ was summarized as the proportional reduction in compound random-effect variance relative to the hierarchy-only model.

For conventional QSPR comparison, measurements were collapsed to a mean log permeability per compound and molecular descriptors were summarized by their compound median. This provided a conventional compound-level benchmark while keeping the molecular information matched as closely as possible to the hierarchical models.

HuSkinDB replication used S1 only because molecular weight and logP were complete for all records, whereas donor temperature was missing in 155 of 546 measurements and the broader Cheruvu descriptor panel was not available in directly comparable form. Temperature was therefore evaluated as a secondary complete-case analysis in HuSkinDB rather than as a required term in the primary HuSkin QSPR.

Accordingly, variance decompositions from Cheruvu et al. and HuSkinDB were used to assess replication of a substantial source-associated component, not to compare the absolute magnitude of variance components directly between datasets. The Cheruvu decomposition adjusts for temperature as a fixed effect, whereas the primary HuSkinDB decomposition does not.

3.5 Leave-one-compound-out validation and prediction estimands

Prediction was evaluated using leave-one-compound-out (LOCO) validation. In each fold, all measurements of one compound were removed before model fitting and descriptor standardization. Three estimands were distinguished explicitly:

1. **Conventional aggregate QSPR:** prediction from the compound-level regression.
2. **Hierarchical, source-blind:** prediction from the hierarchical molecular fixed effects with all random effects set to zero for the held-out compound.
3. **Hierarchical, known-source:** prediction using the molecular fixed effects plus a source random effect when the held-out measurement belonged to a source represented in the training fold. New source levels received zero effect.

This distinction was critical because a known-source hierarchical predictor has access to information unavailable to the conventional aggregate model. The source-blind prediction therefore served as a direct test of whether hierarchical estimation itself produced a superior transferable molecular function $f(X)$, whereas the known-source prediction quantified the additional value of source context.

Performance was summarized using RMSE, MAE, predictive R^2, and compound-balanced RMSE. Sensitivity analyses in Cheruvu examined exclusion of observations with absolute standardized residuals greater than three, descriptor aggregation by mean rather than median, restriction to multi-reference compounds, and an expanded nonlinear molecular specification. These analyses were secondary and did not alter the prespecified primary specifications.

3.6 Chemical applicability domain

Chemical applicability was assessed using descriptors only, separately from experimental-source applicability. For every LOCO fold, descriptor means, standard deviations, covariance matrices, and univariate ranges were estimated **from the training compounds only**. A held-out compound was considered in-domain when (i) its squared Mahalanobis distance was below the 95th percentile of the appropriate chi-square distribution and (ii) every descriptor remained within the corresponding univariate training range. S1 therefore used a two-dimensional domain and S2 a six-dimensional domain.

For the SkinPiX transfer analysis, applicability was defined relative to the complete HuSkin training descriptor space because HuSkinDB was the deployed reference model and SkinPiX was external. No SkinPiX information was used to define the chemical domain.

3.7 Held-out-source calibration in HuSkinDB

To simulate deployment to an unseen experimental source, each HuSkin reference was held out in turn. The model was fitted to all other references. Compounds present both in the held-out source and in the training data were considered candidate **anchor compounds**. Compounds present in the held-out source but absent from all model-fitting data were reserved as genuinely unseen test compounds.

For an anchor compound i measured in held-out source L, the residual relative to the fitted reference model was

$$r_{iL} = y_{iL} - \hat{\mu}_i,$$

where $\hat{\mu}_i$ included the molecular fixed effect and the known compound-specific random effect estimated from the training sources. Repeated anchor measurements were first averaged within compound. For a panel of k anchors, the raw source offset was the mean anchor residual,

$$\bar{r}_L = \frac{1}{k}\sum_{i=1}^{k} r_{iL}.$$

The estimated source effect was shrunk toward zero using the random-effect and residual variances from the training model,

$$\hat{u}_L = \frac{k\hat{\sigma}_u^2}{\hat{\sigma}_\epsilon^2 + k\hat{\sigma}_u^2}\bar{r}_L.$$

Predictions for unseen compounds in the held-out source were then shifted by $\hat{u}_L$. Anchor-panel sizes from 1 to 5 compounds were evaluated where available. For each panel size, 100 random anchor selections were drawn with a fixed random seed.

Prediction error was first calculated separately within each held-out source. For source s and anchor-selection replicate r, RMSE was calculated across the unseen test measurements from that source. For each anchor-panel size, these source-specific RMSE values were averaged across the 100 anchor selections and then across eligible sources, giving equal weight to each experimental source regardless of the number of test measurements it contributed. We refer to this quantity as the mean source-level RMSE. Percentage improvement from calibration was calculated as

$$100\left(1 - \frac{\overline{\mathrm{RMSE}}_{\mathrm{calibrated}}}{\overline{\mathrm{RMSE}}_{\mathrm{uncalibrated}}}\right).$$

Results were summarized both across all sources eligible for a given panel size and in matched source cohorts supporting the same panel sizes. Because the set of eligible sources changes as the required number of anchors increases, inference about anchor number was based primarily on matched cohorts. The 9-source cohort supporting all panel sizes from 1 to 5 anchors was used for the primary HuSkin anchor-number comparison.

3.8 Falsification by wrong-source calibration

A negative control tested whether improved predictions could arise merely from adding an arbitrary offset. For each held-out HuSkin source and anchor-panel size, the correctly estimated source offset was compared with an offset generated from a **different** held-out source with the same anchor-panel size. The novel-compound test set itself was unchanged. If calibration reflects a source-specific latent component, the matched source offset should improve prediction whereas a borrowed offset should not. Source-matched and different-source calibration were evaluated using the same mean source-level RMSE and percentage-improvement definition described above, with comparisons made within the same eligible source set at each anchor-panel size.

3.9 Cross-dataset transfer from HuSkinDB to SkinPiX

The SkinPiX analysis was designed as a stricter cross-dataset transfer test. A hierarchical S1 model was fitted to HuSkinDB using RDKit molecular weight and logP. SkinPiX was not used for model fitting. Within each SkinPiX source, compounds also present in HuSkinDB were candidate anchors, while SkinPiX-only compounds were the test set. A source was eligible when it contained at least one anchor, at least one SkinPiX-only compound, and no DOI overlap with HuSkinDB.

Cross-dataset calibration performance was summarized using the same source-balanced procedure as in the HuSkin held-out-source experiment. RMSE was calculated separately for the SkinPiX-only test measurements within each source and each anchor-selection replicate, averaged across the 100 anchor selections, and then averaged across sources. Thus, each SkinPiX source contributed equally to the reported mean source-level RMSE regardless of its number of test measurements.

The same empirical-Bayes calibration procedure used in the HuSkin internal experiment was then applied to each eligible SkinPiX source. Anchor panels were evaluated where possible with 100 random anchor selections. Because only 4 SkinPiX sources simultaneously contained anchors and genuinely novel compounds, the one-anchor result across these sources was treated as the primary external-transfer result. A matched 3-source subset supporting up to 3 anchors was used only as a secondary anchor-number sensitivity analysis. Larger anchor panels involved progressively fewer sources and were treated descriptively rather than as evidence of a general dose-response.

For compound-level visualization of the one-anchor experiment, calibrated predictions were averaged across the 100 random one-anchor selections for each measurement. This visualization therefore represents expected performance over random selection of one available anchor rather than a single privileged anchor choice.

3.10 Uncertainty estimation

Uncertainty was estimated at the level corresponding to the independent validation unit. For leave-one-compound-out analyses, 95% confidence intervals were obtained by paired cluster bootstrap of held-out compounds. Compounds were sampled with replacement, with all measurements from each selected compound retained together, and RMSE was recalculated from the existing out-of-fold predictions. The same bootstrap sample was used for all prediction approaches within each replicate, preserving the paired comparison. Relative RMSE improvement was calculated against the conventional aggregate QSPR within each bootstrap sample. 2000 bootstrap replicates were used, with percentile intervals defined by the 2.5th and 97.5th percentiles.

Uncertainty in variance partitioning was estimated by parametric bootstrap of the final REML mixed-effects models. For each simulated dataset, the model was refitted and the compound, source, and residual variance components were normalized by their sum. Ninety-five percent percentile intervals were calculated from 2000 bootstrap replicates.

For the matched 9-source HuSkin calibration analysis, uncertainty in mean source-level RMSE improvement was estimated by nonparametric bootstrap of literature sources. 9 sources were sampled with replacement in each replicate, with the same sampled source set retained across anchor-panel sizes. Improvement was recalculated using the same ratio of mean calibrated to mean uncalibrated source-level RMSE defined above.

For the wrong-source falsification analysis, sources were similarly bootstrapped within each anchor-panel size. Matched-source and different-source calibration were evaluated on the same resampled sources, allowing direct estimation of the paired difference in mean source-level RMSE.

Formal confidence intervals were not calculated for the external SkinPiX result because only 4 independent eligible sources were available. External-transfer heterogeneity was instead summarized using individual source results and the number of sources and compounds showing lower error after calibration.

3.11 Software and reproducibility

All statistical analyses were performed in R. Linear mixed-effects models were fitted with `lme4` (Bates et al. 2015). RDKit was used in a separate Python environment to calculate molecular weight and Crippen logP from SMILES (Landrum et al. 2026). Analysis scripts were organized sequentially so that data preparation, model fitting, validation, falsification, and manuscript output were separated and reproducible. No results were recalculated in the manuscript-generation scripts.

4 Results

4.1 Source-associated heterogeneity replicates across datasets

Adding a literature-source random intercept to the Cheruvu hierarchy reduced AIC from 1033.64 to 823.29. Adding temperature further reduced AIC to 812.32, whereas adding skin thickness increased AIC slightly

to 813.66. The temperature coefficient was 0.0425 $\log_{10}$ units per °C ($p =< 0.001$), corresponding to approximately a 10.3% multiplicative increase in K_p per degree Celsius. Skin thickness was not supported ($p = 0.42$). The M2 model was therefore used as the primary Cheruvu hierarchy.

In the REML variance decomposition, the modeled random/residual variance in Cheruvu et al. was partitioned into compound-associated (66.4%, 95% CI 56.9–74.7%), source-associated (22.8%, 14.4–31.9%), and residual (10.8%, 8.3–14.0%) components (Table 2; Figure 1). HuSkinDB independently reproduced substantial source-associated variation: compound, source, and residual components accounted for 23.5% (15.2–32.9%), 44.4% (33.5–54.5%), and 32.1% (24.7–40.8%), respectively. Because the Cheruvu model included temperature as a fixed effect whereas the primary HuSkinDB model did not, these percentages should not be interpreted as a direct quantitative comparison between datasets. Rather, the replicated result is that a substantial source-associated component was present in both datasets.

Table 2: Partitioning of modeled random/residual variance. Percentages are normalized over the compound, source, and residual variance components; fixed-effect variation is not included in the denominator. The Cheruvu and HuSkinDB decompositions use different fixed-effect specifications and should therefore be interpreted as independent evidence of substantial source-associated variation rather than compared directly in magnitude.

Dataset	Variance component	Estimate (%)	95% CI
Cheruvu	Compound	66.4	[56.9, 74.7]
Cheruvu	Source	22.8	[14.4, 31.9]
Cheruvu	Residual	10.8	[8.3, 14.0]
HuSkinDB	Compound	23.5	[15.2, 32.9]
HuSkinDB	Source	44.4	[33.5, 54.5]
HuSkinDB	Residual	32.1	[24.7, 40.8]

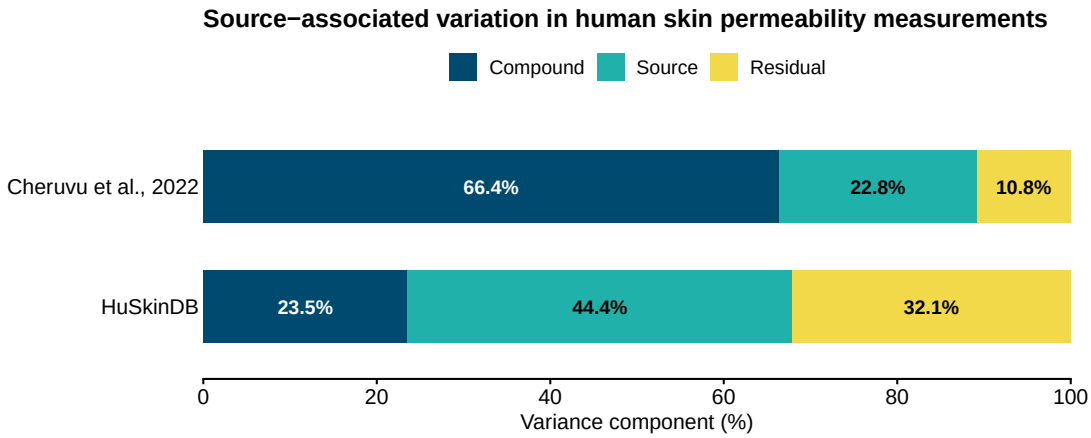


Figure 1: Variance partitioning in Cheruvu et al. 2022 and HuSkinDB. Bars show the relative contributions of compound, literature source, and residual components to modeled random/residual variance. The Cheruvu decomposition adjusts for temperature, whereas the primary HuSkinDB decomposition does not because temperature was incomplete; the figure therefore demonstrates replication of substantial source-associated variation rather than a direct comparison of component magnitudes between datasets.

4.2 Molecular descriptors explain substantial compound variation, but source context drives most of the predictive gain

In Cheruvu, S1 (molecular weight plus logKow) explained 60.3% of the compound random-effect variance. The broader S2 specification explained 61.4%, only 1.1 percentage points more. Thus, most of the explainable compound signal captured by these prespecified models was already present in the simple size–lipophilicity specification. In HuSkinDB, molecular weight plus logP explained 37.1% of between-compound variance.

LOCO validation separated improvement in the molecular component from improvement due to source context. Relative to conventional aggregation, source-blind hierarchical prediction changed RMSE by 5.4% (95% CI -3.6–10.9%) for Cheruvu S1, 7.0% (-5.1–14.0%) for Cheruvu S2, and -1.7% (-5.1–1.0%) for HuSkinDB S1. All three intervals included zero, providing no evidence that hierarchical fitting alone consistently improved the transferable molecular predictor.

In contrast, known-source hierarchical prediction improved RMSE by 16.7% (95% CI 11.6–21.8%) for Cheruvu S1, 18.7% (11.5–23.3%) for Cheruvu S2, and 17.0% (10.0–23.8%) for HuSkinDB S1. The intervals excluded zero in all three analyses, showing that the reproducible predictive benefit arose primarily when source-specific information was available (Table 3; Figure 2).

Table 3: Leave-one-compound-out prediction under conventional, source-blind hierarchical, and known-source hierarchical settings. Confidence intervals were obtained by paired compound-level bootstrap resampling of held-out compounds.

Dataset	Specification	Prediction setting	RMSE (95% CI)	RMSE improvement vs conventional (%)
Cheruvu	S1	Conventional aggregate QSPR	0.866 [0.725, 1.017]	Reference
Cheruvu	S1	Hierarchical, source blind	0.819 [0.708, 0.945]	5.4 [-3.6, 10.9]
Cheruvu	S1	Hierarchical, known source	0.721 [0.601, 0.855]	16.7 [11.6, 21.8]
Cheruvu	S2	Conventional aggregate QSPR	0.850 [0.715, 0.977]	Reference
Cheruvu	S2	Hierarchical, source blind	0.790 [0.703, 0.883]	7.0 [-5.1, 14.0]
Cheruvu	S2	Hierarchical, known source	0.691 [0.602, 0.788]	18.7 [11.5, 23.3]
HuSkinDB	S1	Conventional aggregate QSPR	1.333 [1.198, 1.474]	Reference
HuSkinDB	S1	Hierarchical, source blind	1.355 [1.201, 1.537]	-1.7 [-5.1, 1.0]
HuSkinDB	S1	Hierarchical, known source	1.106 [0.959, 1.305]	17.0 [10.0, 23.8]

4.3 Shared compounds calibrate completely held-out HuSkin sources

Holding out entire HuSkin references tested whether source information could be recovered rather than assumed known. Calibration used only compounds shared between the held-out source and the training corpus; evaluation was restricted to compounds from that source that were absent from all model-fitting data.

Calibration improved mean source-level RMSE at every tested anchor-panel size among eligible sources. Because eligibility changed with panel size, the primary anchor-number comparison used the same 9 references

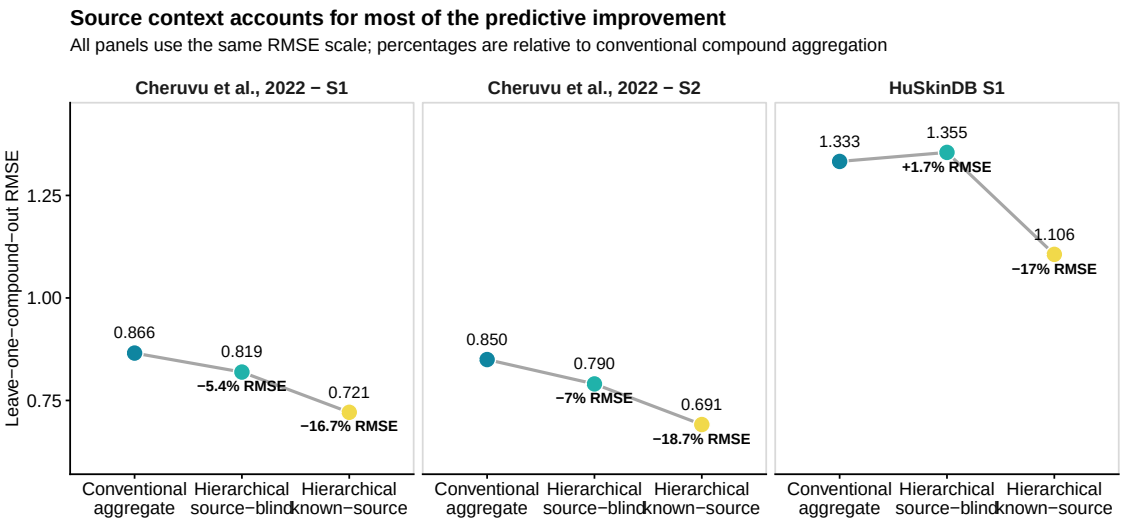


Figure 2: Leave-one-compound-out RMSE under conventional aggregate QSPR, source-blind hierarchical prediction, and hierarchical prediction with known source context. Common y-axis limits show that source-blind gains were modest in Cheruvu et al. and absent in HuSkinDB, whereas source context produced the larger and reproducible reduction in prediction error.

that supported 1 through 5 anchors. Within this matched cohort, mean source-level RMSE improvement was 31.1% (95% CI 13.4–42.1%) with 1 anchor and 45.3% (19.2–60.0%) with 5 anchors. Point estimates increased monotonically with anchor-panel size, although the overlapping intervals do not support interpreting each incremental increase as statistically distinct.

The cohort-level trend did not imply universal source-level success. At 5 anchors, 8 of 9 matched sources improved, while HUR_48 deteriorated substantially. The median source-specific RMSE improvement at 5 anchors was 40.0%, whereas the reduction in mean source-level RMSE was 45.3%. These quantities differ because the latter is calculated from the ratio of the mean calibrated and uncalibrated source-level RMSEs, rather than as the arithmetic mean or median of source-specific percentage improvements.

Table 4: Held-out HuSkin source calibration in the matched-source cohort. Confidence intervals were obtained by bootstrap resampling of literature sources.

Anchors	Matched sources	RMSE uncalibrated	RMSE calibrated	RMSE improvement (%)	95% CI
1	9	1.135	0.782	31.1	[13.4, 42.1]
2	9	1.135	0.705	37.9	[16.0, 51.0]
3	9	1.135	0.675	40.5	[17.2, 54.8]
4	9	1.135	0.641	43.6	[18.4, 57.8]
5	9	1.135	0.621	45.3	[19.2, 60.0]

4.4 Wrong-source calibration falsifies a generic-offset explanation

The wrong-source negative control produced the opposite behavior from source-matched calibration. Source-matched and different-source percentage improvements are reported for each anchor-panel size in Table 4B. Because the eligible source cohort changed with panel size, these values should be compared within rather than across anchor-panel sizes.

More importantly, the paired difference between different-source and matched-source RMSE was positive at every anchor-panel size. It was 0.225 (95% CI 0.117–0.343) with 1 anchor and 0.631 (0.390–0.936) with 5 anchors, with all intermediate confidence intervals also remaining above zero. Thus, calibration using source-matched information consistently outperformed an otherwise equivalent offset borrowed from another source (Table 4B; Figure 4). This result makes arbitrary recentering and regression-to-the-mean explanations unlikely: the useful calibration signal had to be associated with the same experimental source as the target measurements.

Table 5: Falsification control comparing source-matched calibration with an offset borrowed from a different held-out HuSkin source.

Anchors	Eligible sources	Matched-source RMSE improvement (%)	Different-source RMSE improvement (%)
1	31	9.3	-9.0
2	22	15.8	-10.9
3	19	21.0	-15.8
4	12	36.2	-16.1
5	9	45.3	-10.3

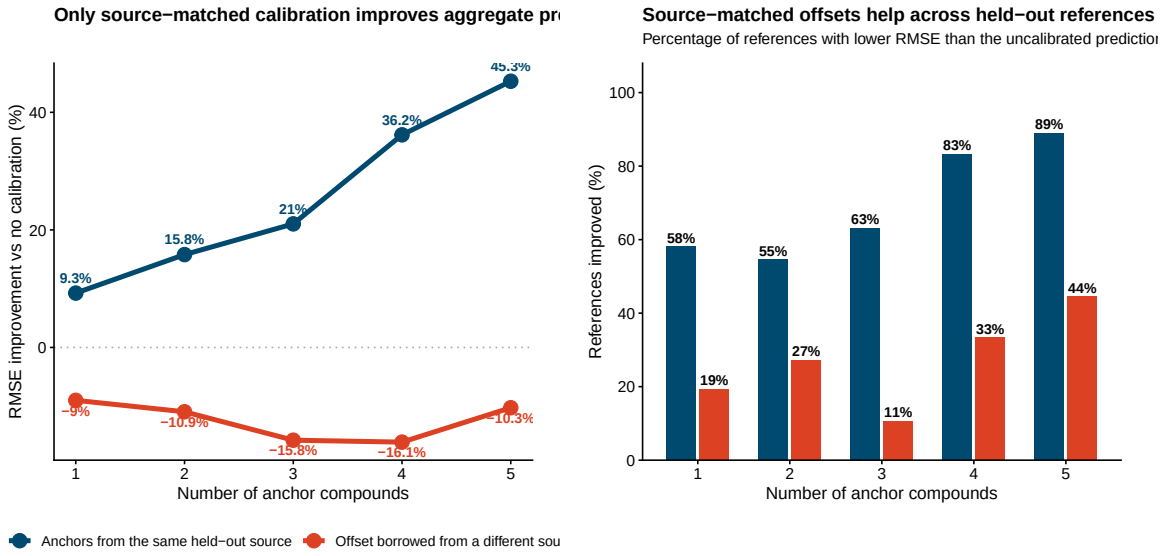


Figure 3: Falsification of a generic-offset explanation. Panel A shows mean source-level RMSE improvement when calibration is estimated from anchors measured in the same held-out source versus an offset borrowed from a different source. Panel B shows the proportion of held-out references improved by each procedure. Source-matched calibration helps; mismatched source information reverses the benefit.

For the compound-level visualization of the one-anchor experiment, calibrated predictions were first averaged across the 100 random one-anchor selections for each test measurement. Absolute errors were then calculated from these mean predictions and averaged within compound where necessary. This compound-level summary was used only for visualization and for counting compounds with lower absolute error after calibration; it is distinct from the mean source-level RMSE used for the primary calibration-performance estimate.

4.5 HuSkin-trained calibration transfers to independent SkinPiX sources

The cross-dataset analysis tested the same mechanism under a stricter separation between model development and deployment. The molecular model was fitted to 546 HuSkin measurements from 251 compounds. SkinPiX contained compounds overlapping HuSkinDB and SkinPiX-only compounds as documented in the transfer audit. Of the sources containing HuSkin-overlapping compounds, 4 contained both an available anchor and at least one SkinPiX-only compound suitable for genuinely external evaluation. Across these sources, the test set comprised 54 unique compounds not present in HuSkinDB.

With one randomly selected anchor compound per source, calibration reduced mean source-level RMSE by 17.6% across the 4 eligible SkinPiX sources. Because RMSE was calculated within source and then averaged across sources, each source contributed equally to this estimate regardless of its number of test measurements. At the source level, 3 of 4 sources improved. At the compound level, 44 of 54 unseen SkinPiX compounds had lower mean absolute error after calibration when predictions were averaged across the 100 one-anchor selections. The source-level RMSE and compound-level absolute-error summaries therefore describe complementary but distinct aspects of transfer performance.

3 of the 4 eligible SkinPiX sources supported at least 3 anchors. Restricting the analysis to this matched subset, mean source-level RMSE improvement increased across the supported anchor-panel sizes, from 18.6% at the smallest panel to 23.9% at the largest. This small matched subset provides supportive rather than definitive evidence for an anchor-number effect across databases. Estimates based on still larger anchor requirements involved progressively fewer sources and were not emphasized.

All 54 external SkinPiX test compounds were inside the HuSkin-defined chemical applicability domain. The transfer result should therefore be interpreted primarily as **experimental-source transfer within HuSkin chemical space**, not as evidence for reliable chemical extrapolation beyond that space.

Table 6: Cross-dataset calibration of SkinPiX sources using a model trained only on HuSkinDB.

Cohort	Anchors	Sources	RMSE uncalibrated	RMSE calibrated	RMSE improvement (%)
All eligible sources	1	4	0.810	0.667	17.6
Matched three-source cohort	1	3	0.949	0.772	18.6
Matched three-source cohort	2	3	0.949	0.740	22.1
Matched three-source cohort	3	3	0.949	0.722	23.9

4.6 Applicability-domain diagnostics

Applicability-domain analysis was used as a deployment qualification rather than as an additional model-selection criterion. Cheruvu and HuSkinDB applicability-domain counts are reported in Supplementary Table S7. All 54 SkinPiX compounds used in the external transfer test were in-domain relative to the HuSkin training set. Domain-stratified errors and normalized fold-specific Mahalanobis distances are reported in Supplementary Table S7 and Supplementary Figure S1.

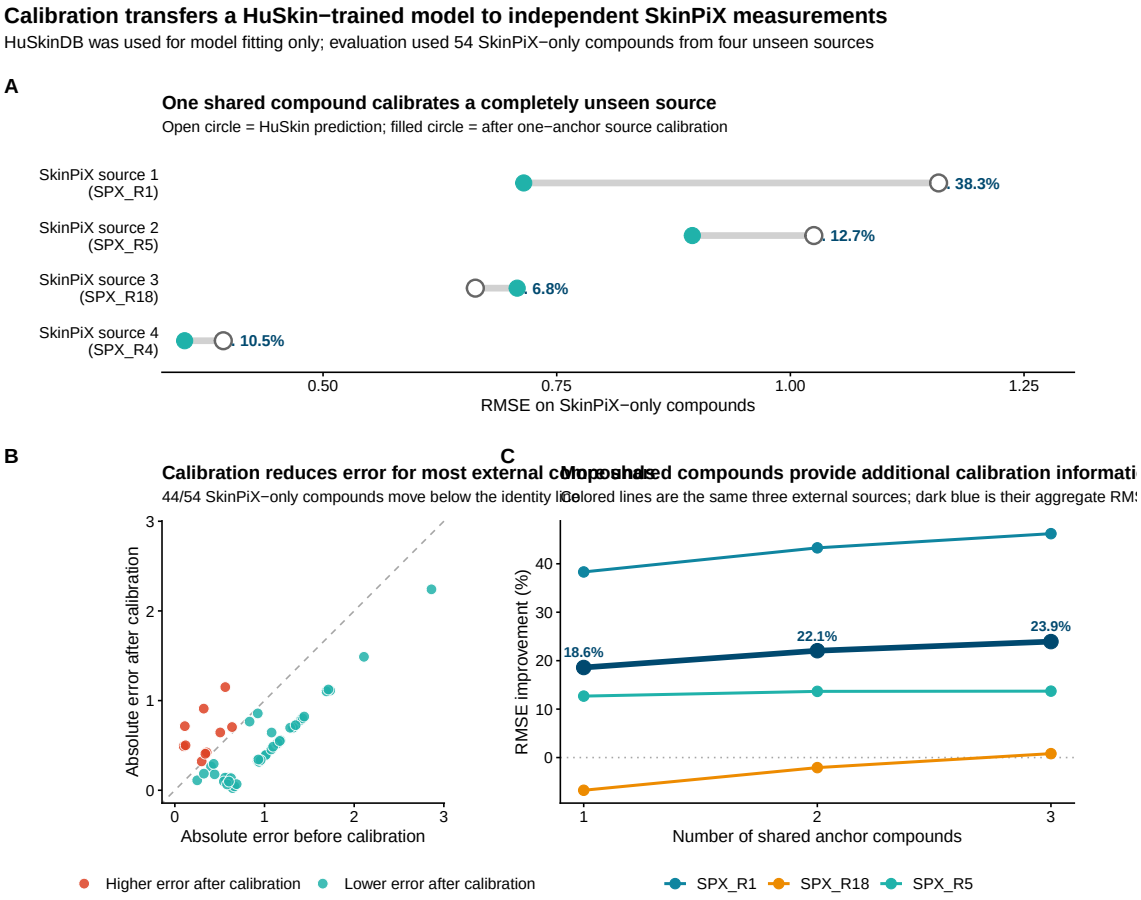


Figure 4: Cross-dataset transfer from HuSkinDB to SkinPiX. Panel A compares uncalibrated and one-anchor calibrated RMSE within each eligible unseen SkinPiX source. Panel B compares compound-level absolute prediction error before and after calibration for SkinPiX-only compounds. Panel C shows the matched source subset supporting multiple anchors; colored lines represent individual sources and the dark line represents reduction in mean source-level RMSE across those same sources.

5 Discussion

5.1 Source-associated heterogeneity is a reproducible part of permeability data

The central result is not simply that a mixed-effects model fits the literature better. Rather, two independently curated datasets show that a substantial component of measurement variability is associated with literature source, and that this source-associated component carries predictive information across compounds measured within the same source. Source-associated variation represented 22.8% (95% CI 14.4–31.9%) of modeled random/residual variance in Cheruvu et al. and 44.4% (33.5–54.5%) in HuSkinDB. Because the two decompositions were not based on identical fixed-effect specifications, their numerical magnitudes should not be compared directly. The important replicated finding is that source-associated variation constituted a substantial component in both independently curated datasets.

This result is consistent with decades of evidence that permeability depends on experimental context (Fujiwara et al. 2003; Abraham and Martins 2004; Samaras et al. 2012). The present analysis does not claim that heterogeneity is newly discovered. Its contribution is to show that, under a simple hierarchical model, part of that heterogeneity behaves like a **calibratable source-level quantity** that can be estimated from shared compounds and then transferred to compounds not used in fitting.

5.2 Hierarchical fitting does not automatically create a better universal molecular QSPR

The source-blind analysis narrows the interpretation of the LOCO results. Although the point estimates in Cheruvu suggested lower RMSE under source-blind hierarchical fitting, the corresponding bootstrap intervals included no improvement. HuSkinDB similarly showed no source-blind benefit. In contrast, the larger reductions obtained when source context was available had confidence intervals entirely above zero in both development and replication datasets. The evidence therefore does not support hierarchical fitting as a consistently superior estimator of the universal molecular function $f(X)$; the reproducible predictive gain arises from representing source-specific information.

This is scientifically useful because it argues against framing the method as a universally superior estimator of $f(X)$. The data instead support a two-part prediction problem. Molecular descriptors determine the expected permeability of an unseen compound, while source information shifts that expectation toward the scale of a particular experimental system. The value of the hierarchical formulation is therefore not that it always discovers a better molecular equation, but that it separates molecular prediction from source calibration.

The modest difference between Cheruvu S1 and S2 reinforces this point. Expanding from molecular weight and logKow to the broader S2 descriptor set increased explained compound variance by only 1.1 percentage points. A more complex descriptor search might improve point prediction, but it would obscure the methodological question addressed here and increase the risk that performance differences were driven by feature selection rather than by the treatment of measurement hierarchy.

5.3 Anchor compounds convert an unseen source from unknown to calibratable

The held-out-source experiment is closest to the intended deployment setting. A new source initially has no estimated random effect, so prediction defaults to the reference scale. Once shared compounds are measured, their residuals relative to the reference model provide information about the new source's systematic displacement. Empirical-Bayes shrinkage prevents a small anchor panel from being interpreted as a perfectly known offset.

The matched 9-source HuSkin analysis showed a monotonic point-estimate improvement as anchors increased from 1 to 5. Bootstrap intervals remained above zero at every anchor-panel size, including 31.1% (95% CI 13.4–42.1%) with 1 anchor and 45.3% (19.2–60.0%) with 5 anchors. However, the substantial overlap between panel-size intervals means that the monotonic point-estimate trend should not be interpreted as

evidence that each additional anchor confers a statistically distinct benefit. HUR_48 also demonstrates that a single source-wide intercept is not universally adequate. Possible explanations include compound-by-protocol interactions, condition changes within a publication, incorrect metadata grouping, formulation effects, or other deviations from the common-offset assumption. That failure case should be treated as part of the method's operating characteristics rather than removed post hoc.

Most importantly, the wrong-source control strongly supports the proposed mechanism. The paired falsification contrast was positive at every anchor-panel size, with all bootstrap confidence intervals for different-source minus matched-source RMSE remaining above zero. This comparison is stronger than considering the two calibration procedures independently because both were evaluated on the same resampled target sources. If calibration improvement arose from arbitrary recentering, flexible model fitting, or regression toward an overall mean, an offset drawn from another source should frequently provide similar benefit. Instead, useful calibration information was specific to the experimental source being calibrated.

5.4 Cross-dataset transfer provides the strongest deployment-oriented evidence

SkinPiX was intentionally withheld from model development. HuSkinDB provided the molecular model, SkinPiX-overlapping compounds supplied anchors, and SkinPiX-only compounds supplied the test set. No DOI overlap was identified between the two source sets used for transfer, and RDKit descriptors were recalculated from structure in both databases to avoid descriptor-definition mismatch.

The resulting one-anchor improvement of 17.6% across 4 SkinPiX sources is therefore qualitatively different from an internal cross-validation gain. It shows that a source-calibration procedure developed and validated inside one database can improve predictions in another independently curated database. The result should nevertheless be interpreted with the correct level of confidence. Only 4 SkinPiX sources were eligible for true anchor-plus-novel-compound testing, and 1 did not improve with one-anchor calibration. The external evidence is therefore promising and mechanistically consistent with the HuSkin experiment, but not yet large enough to establish the frequency with which calibration will succeed in arbitrary laboratories or protocols. Formal confidence intervals were not calculated for this external-transfer estimate because only 4 independent eligible SkinPiX sources were available; treating the 54 compounds as independent external validation units would overstate the effective replication.

The fact that all 54 external test compounds were inside the HuSkin chemical domain is both a strength and a limitation. It strengthens the interpretation that the experiment tests **source transfer** rather than confounding source transfer with severe chemical extrapolation. Conversely, it means the present study provides no evidence that source calibration rescues predictions for compounds outside the molecular domain of the reference model. Chemical applicability and experimental-source applicability should therefore remain separate dimensions of deployment confidence.

5.5 Relation to previous work

Previous approaches have addressed permeability heterogeneity through standardization, latent common permeability, explicit experimental covariates, filtering, aggregation, and more recent record-level or mixed-effects analyses (Fujiwara et al. 2003; Abraham and Martins 2004; Samaras et al. 2012; Oh and Ahn 2026). The present framework is complementary rather than contradictory. Instead of attempting to remove all source variation before modeling, it retains measurement-level structure and asks whether the residual source component can be used operationally.

This distinction also separates the current analysis from studies whose primary objective is to maximize predictive performance from merged databases (Asgarkhanova et al. 2026; Oh and Ahn 2026). Pooling HuSkinDB and SkinPiX at the outset could produce a larger training set, but it would eliminate the clean cross-dataset transfer experiment. By preserving SkinPiX as an external source collection, the analysis tests a deployment behavior that would otherwise be difficult to identify.

5.6 Limitations

Several limitations constrain interpretation. First, the random reference effect is **source-associated**, not a causal estimate of laboratory, apparatus, operator, skin donor, or protocol effects individually. A publication-level intercept can absorb all systematic differences correlated with that source. The present results therefore do not justify attributing a fixed percentage of permeability variation causally to apparatus differences.

Second, a common random intercept assumes that a source shifts compounds approximately in parallel. The HUR_48 failure shows that this assumption can be violated. Future models could examine random slopes, protocol interactions, or hierarchical decomposition of source into laboratory, apparatus, tissue preparation, and other metadata where sufficiently complete data exist. Such extensions should be justified by new data rather than added solely to improve retrospective fit.

Third, the external SkinPiX validation is limited by source overlap. Although 54 novel compounds were evaluated, they came from only 4 eligible SkinPiX sources. The number of experimental sources, rather than the number of compound measurements alone, is therefore the main limitation on the generality of the cross-dataset conclusion.

Fourth, the chemical component was intentionally simple. The objective was to isolate the effect of measurement hierarchy and source calibration while holding molecular information relatively fixed, not to establish the highest-performing skin-permeability predictor available. More expressive molecular models could be substituted for $f(X)$, but they would require separate validation and should not be assumed to preserve the same calibration behavior.

Finally, the analyses are retrospective and rely on literature sources. The strongest validation would be prospective: fit and freeze the reference model, measure several chemically diverse anchor compounds on a previously unseen experimental platform, estimate its platform offset, and then predict additional compounds before observing their permeability. Such a study would directly test whether literature-derived source calibration transfers to a real new apparatus rather than another historical database.

5.7 Implications for experimental deployment

The framework suggests a practical deployment strategy. A reference model provides a molecular prediction on a common permeability scale. A new laboratory or platform then measures a small panel of anchor compounds already represented in the reference corpus. Their deviations from the reference expectation estimate a platform-specific offset, after shrinkage for limited anchor information. Predictions for new compounds can then be reported together with two separate applicability statements: whether the compound is chemically inside the reference domain and whether the experimental platform has been calibrated.

This creates four qualitatively different deployment states: (i) chemically in-domain and platform-calibrated, (ii) chemically in-domain but platform-uncalibrated, (iii) chemically out-of-domain but platform-calibrated, and (iv) chemically out-of-domain and platform-uncalibrated. The present study provides the strongest evidence for the first state and does not imply that source calibration compensates for chemical extrapolation.

6 Conclusions

Human skin permeability measurements contain substantial and reproducible source-associated variation that is not captured by molecular descriptors alone. Retaining measurement-level hierarchy does not consistently produce a better source-independent molecular QSPR, but it allows source context to be represented explicitly. When a source is known, this context materially improves prediction; when a source is new, shared anchor compounds can estimate a useful source offset.

The calibration effect survived two important tests. It improved prediction of compounds absent from model fitting when entire HuSkin sources were held out, and it failed when the offset was deliberately borrowed from the wrong source. The same approach then transferred from HuSkinDB to independent SkinPiX

measurements, improving one-anchor prediction across 54 HuSkin-unseen compounds from 4 eligible sources. These findings support a study-aware QSPR framework in which molecular prediction and experimental-source calibration are treated as distinct but complementary components of permeability prediction.

7 Data and code availability

The analysis uses publicly available datasets from Cheruvu et al. 2022, HuSkinDB, and SkinPiX (Cheruvu et al. 2022; Stepanov et al. 2020; Chedik et al. 2024). Analysis scripts and processed outputs are maintained in the project repository. The final repository release should include the exact raw-data provenance, software-environment information for RDKit descriptor generation, and all scripts required to regenerate manuscript tables and figures.

8 Author contributions

Chayada Piantham: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review and editing.

Additional author contributions should be added according to the final author list and CRediT roles.

9 Funding

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10 Competing interests

The author(s) declare no competing interests unless otherwise stated at submission.

11 Supplementary material

11.1 Supplementary Table S1. Cheruvu hierarchical model comparison

Table 7: Cheruvu hierarchical model comparison.

model	AIC	BIC	logLik	df
M0	1033.6365	1046.1328	-513.8183	3
M1	823.2880	839.9497	-407.6440	4
M2	812.3239	833.1510	-401.1620	5
M3	813.6605	838.6530	-400.8302	6

11.2 Supplementary Table S2. Molecular QSPR specifications

Table 8: QSPR model summary and reduction in compound variance.

specification	n_compounds	n_observations	compound_variance	compound_baseline_variance	variance_reduction	joint_AIC	baseline_AIC	AIC_reduction	joint_BIC	baseline_BIC	BIC_reduction	LRT_stat	LRT_df	LRT_p
S1	148	476	0.6911021	0.2743721	60.29933	812.3239	704.5228	83.151	733.6791	911.8020	2	0		
S2	148	476	0.6911021	0.2664700	61.44274	812.3239	703.9748	83.151	749.7944	920.3491	6	0		

Table 9: QSPR coefficient estimates.

specification	model	term	estimate	se
S1	Conventional	(Intercept)	-2.4415490	0.0628978
S1	Conventional	mw_z	-0.5250908	0.0643767
S1	Conventional	log_kow_z	0.6355819	0.0643767
S1	Hierarchical	(Intercept)	-2.6123979	0.0830520
S1	Hierarchical	temperature_c	0.0380659	0.0112105
S1	Hierarchical	mw_z	-0.3697536	0.0584279
S1	Hierarchical	log_kow_z	0.6367630	0.0536842
S2	Conventional	(Intercept)	-2.4415490	0.0594157
S2	Conventional	mw_z	-0.4497299	0.1426366
S2	Conventional	log_kow_z	0.3260018	0.1201162
S2	Conventional	melting_point_k_z	-0.2378772	0.0754869
S2	Conventional	log_saq_z	-0.2132459	0.1263555
S2	Conventional	h_donor_z	-0.2450708	0.0943533
S2	Conventional	h_acceptor_z	0.0730053	0.1317609
S2	Hierarchical	(Intercept)	-2.6044158	0.0819660
S2	Hierarchical	temperature_c	0.0380549	0.0111729
S2	Hierarchical	mw_z	-0.1451807	0.1266952
S2	Hierarchical	log_kow_z	0.5000372	0.1008058
S2	Hierarchical	melting_point_k_z	-0.1557432	0.0661164
S2	Hierarchical	log_saq_z	0.0008715	0.1059463
S2	Hierarchical	h_donor_z	-0.0400587	0.0841324
S2	Hierarchical	h_acceptor_z	-0.1463498	0.1117261

11.3 Supplementary Table S3. Cheruvu validation, applicability, and sensitivity analyses

Table 10: Cheruvu leave-one-compound-out performance.

specification	approach	RMSE	MAE	R2	r
S1	Conventional	0.8655258	0.6904392	0.3886059	0.6304196
S1	Hierarchical	0.7210606	0.5567135	0.5756689	0.7609185
S2	Conventional	0.8497195	0.6743949	0.4107326	0.6494215
S2	Hierarchical	0.6912306	0.5338598	0.6100516	0.7836126

Table 11: Cheruvu applicability-domain performance summary.

specification	in_domain	n_compounds	MAE_conventional	MAE_hierarchical
S1	FALSE	11	1.0648280	0.9484063
S2	FALSE	20	0.9834190	0.8814771
S1	TRUE	137	0.6216955	0.5062083
S2	TRUE	128	0.5780136	0.4843456

Table 12: Cheruvu sensitivity analyses.

scenario	specification	approach	RMSE	MAE	R2	r
Primary S2	S2	Conventional	0.8497195	0.6743949	0.4107326	0.6494215
Primary S2	S2	Hierarchical	0.6912306	0.5338598	0.6100516	0.7836126
Multi-reference compounds	Multi-reference compounds	Conventional	0.9520605	0.7368337	0.2865776	0.6742545
Multi-reference compounds	Multi-reference compounds	Hierarchical	0.7612102	0.6180607	0.5439348	0.7701428
Exclude |standardized residual| > 3	Exclude |standardized residual| > 3	Conventional	0.8395476	0.6678751	0.4210067	0.6567405
Exclude |standardized residual| > 3	Exclude |standardized residual| > 3	Hierarchical	0.6751499	0.5229009	0.6255591	0.7919347
Mean descriptor aggregation	Mean descriptor aggregation	Conventional	0.8504152	0.6749500	0.4097673	0.6486841
Mean descriptor aggregation	Mean descriptor aggregation	Hierarchical	0.6912306	0.5338598	0.6100516	0.7836126
Expanded nonlinear specification	Expanded nonlinear specification	Conventional	0.7325140	0.5597719	0.5620817	0.7501002
Expanded nonlinear specification	Expanded nonlinear specification	Hierarchical	0.6713274	0.5189553	0.6321845	0.8013794

11.4 Supplementary Table S4. HuSkinDB replication details

Table 13: HuSkinDB replication summary.

n_observations	n_compounds	n_references	AIC_conventional	AIC_hierarchical	conventional_compound_reference_variance	hierarchical_compound_reference_variance	residual_variance	conventional_pseudo_R2	hierarchical_pseudo_R2	delta_AIC_hierarchical_minus_conventional	AIC_favors_hierarchical	explained_variance_ratio_hierarchical_over_conventional	compound_reference_variance_ratio_hierarchical_over_conventional	replication_supported
546	251	95	1819.391	1622.438	23.53883	44.38055	32.08062	37.11018	-40.34403	391	-0.0541964	0.038257	1.886	FALSE

11.5 Supplementary Table S5. HuSkin calibration across all eligible sources

Table 14: HuSkin held-out-source calibration across all sources eligible at each anchor-panel size.

task	anchors	RMSE_uncalibrated	RMSE_calibrated	MAE_uncalibrated	MAE_calibrated	RMSE_improvement_percent	MAE_improvement_percent	n_references
Known compound	1	0.9733927	0.8868665	0.8796340	0.8012970	8.889135	8.905639	43
Known compound	2	0.9195535	0.7765842	0.8311594	0.7063222	15.547686	15.019655	32
Known compound	3	1.0079967	0.7801664	0.9052829	0.6982215	22.602287	22.872560	22
Known compound	4	1.0670644	0.7372717	0.9618126	0.6514099	30.906539	32.272685	17
Known compound	5	1.1193718	0.7274227	1.0323636	0.6558016	35.015094	36.475711	14
New compound	1	1.2345701	1.1245475	1.1557362	1.0384358	8.911814	10.149416	31
New compound	2	1.1659703	0.9819211	1.0949088	0.8984951	15.785064	17.938823	22
New compound	3	1.0013440	0.7955098	0.9207253	0.7206778	20.555790	21.727167	19
New compound	4	1.0955663	0.7008523	1.0062998	0.6360499	36.028309	36.793203	12
New compound	5	1.1353881	0.6210905	1.0493962	0.5682710	45.297075	45.847813	9

11.6 Supplementary Table S6. HuSkin matched-source calibration by reference

Table 15: Source-level HuSkin calibration results for the matched-source cohort.

reference_id	anchors	RMSE_uncalibrated	RMSE_calibrated	MAE_uncalibrated	MAE_calibrated	RMSE_improvement_percent	MAE_improvement_percent
HUR_10	1	0.7165456	0.6966213	0.5884309	0.6010274	2.780606	-2.1406975
HUR_11	1	1.4441848	0.9867621	1.4149938	0.9424524	31.673416	33.3952978
HUR_14	1	0.5096445	0.4403946	0.3891261	0.3810407	13.587870	2.0778213
HUR_34	1	1.0615460	0.8435379	1.0615460	0.8435379	20.536851	20.5368511
HUR_39	1	1.0499628	0.6520381	0.9903067	0.5790930	37.898930	41.5238733
HUR_48	1	0.6001922	0.7199398	0.4292958	0.6382168	-19.951538	-48.6659731
HUR_5	1	2.6411893	1.2761956	2.5490880	1.0529894	51.681024	58.6915235
HUR_62	1	1.3474212	0.7402004	1.3204707	0.6881406	45.065403	47.8867184

reference_id	anchors	RMSE_uncalibrated	RMSE_calibrated	MAE_uncalibrated	MAE_calibrated	RMSE_improvement_percent	MAE_improvement_percent
HUR_82	1	0.8478066	0.6816903	0.7013074	0.5904566	19.593656	15.8063001
HUR_10	2	0.7165456	0.6769146	0.5884309	0.5863068	5.530839	0.3609709
HUR_11	2	1.4441848	0.8631680	1.4149938	0.8122592	40.231471	42.5962737
HUR_14	2	0.5096445	0.4024432	0.3891261	0.3434788	21.034516	11.7307258
HUR_34	2	1.0615460	0.8178852	1.0615460	0.8178852	22.953395	22.9533950
HUR_39	2	1.0499628	0.6307941	0.9903067	0.5652431	39.922238	42.9224146
HUR_48	2	0.6001922	0.7910804	0.4292958	0.7102240	-31.804523	-65.4393043
HUR_5	2	2.6411893	1.0034004	2.5490880	0.8832719	62.009524	65.3494938
HUR_62	2	1.3474212	0.5731100	1.3204707	0.5142399	57.466156	61.0563180
HUR_82	2	0.8478066	0.5904791	0.7013074	0.5163119	30.352146	26.3786567
HUR_10	3	0.7165456	0.6473512	0.5884309	0.5618797	9.656672	4.5121932
HUR_11	3	1.4441848	0.8047500	1.4149938	0.7504546	44.276517	46.9641075
HUR_14	3	0.5096445	0.3736306	0.3891261	0.3335314	26.687997	14.2870661
HUR_34	3	1.0615460	0.8288580	1.0615460	0.8288580	21.919730	21.9197302
HUR_39	3	1.0499628	0.6333789	0.9903067	0.5646891	39.676065	42.9783652
HUR_48	3	0.6001922	0.8276817	0.4292958	0.7403351	-37.902777	-72.4533767
HUR_5	3	2.6411893	0.9011428	2.5490880	0.7726990	65.881174	69.6872362
HUR_62	3	1.3474212	0.5096748	1.3204707	0.4547776	62.174055	65.5594345
HUR_82	3	0.8478066	0.5514868	0.7013074	0.5036142	34.951343	28.1892353
HUR_10	4	0.7165456	0.6466452	0.5884309	0.5627516	9.755199	4.3640280
HUR_11	4	1.4441848	0.7538536	1.4149938	0.6957659	47.800748	50.8290483
HUR_14	4	0.5096445	0.3665498	0.3891261	0.3300470	28.077351	15.1824990
HUR_34	4	1.0615460	0.7625506	1.0615460	0.7625506	28.166033	28.1660330
HUR_39	4	1.0499628	0.5416488	0.9903067	0.4706125	48.412574	52.4781055
HUR_48	4	0.6001922	0.8720116	0.4292958	0.7755574	-45.288729	-80.6580454
HUR_5	4	2.6411893	0.8311714	2.5490880	0.6913980	68.530413	72.8766523
HUR_62	4	1.3474212	0.4667468	1.3204707	0.4122841	65.359994	68.7774859
HUR_82	4	0.8478066	0.5263467	0.7013074	0.4848802	37.916663	30.8605264
HUR_10	5	0.7165456	0.6386603	0.5884309	0.5587103	10.869554	5.0508202
HUR_11	5	1.4441848	0.7315112	1.4149938	0.6718240	49.347813	52.5210669
HUR_14	5	0.5096445	0.3500772	0.3891261	0.3291176	31.309518	15.4213433
HUR_34	5	1.0615460	0.7635766	1.0615460	0.7635766	28.069383	28.0693826
HUR_39	5	1.0499628	0.5128622	0.9903067	0.4414473	51.154245	55.4231709
HUR_48	5	0.6001922	0.8816582	0.4292958	0.7816612	-46.895975	-82.0798479
HUR_5	5	2.6411893	0.7577549	2.5490880	0.6913980	71.310085	72.8766523
HUR_62	5	1.3474212	0.4447046	1.3204707	0.3973623	66.995871	69.9075284
HUR_82	5	0.8478066	0.5090093	0.7013074	0.4793415	39.961633	31.6503048

11.7 Supplementary Table S7. Applicability-domain summary

Table 16: Chemical applicability-domain summary across development, replication, and transfer analyses.

dataset	specification	n_in_domain	n_out_domain	MAE_in_domain	MAE_out_domain
Cheruvu	S1	137	11	0.506	0.948
Cheruvu	S2	128	20	0.484	0.881
HuSkinDB	S1	243	8	0.807	2.137
SkinPiX external test	S1	54	0	NA	NA

11.8 Supplementary Table S8. SkinPiX overlap and descriptor audit

Table 17: SkinPiX transfer and provenance audit summary.

metric	value
Retained measurements	199
Compounds	108
References	36
HuSkin-overlapping compounds	24
SkinPiX-only compounds	84
Eligible transfer references	4
References with DOI overlap	0
Median absolute RDKit MW difference among matched compounds	0
Median absolute RDKit logP difference among matched compounds	0

Table 18: HuSkin–SkinPiX compound-overlap audit.

compound_id	compound_name	smiles	huskin_compound_id	huskin_compound_name	huskin_smiles	match_method
SPX_C25	2-Nitro-p-phenylenediamine	C1=CC(=C(C=C1N)N+[O-])N	HUC_42	2-Nitro-p-Phenylenediamine	Nc1ccc(c(c1)N+[O-])N	Name
SPX_C30	Benzene	C1=CC=CC=C1	HUC_40	Benzene	c1ccccc1	Name
SPX_C20	Benzoic acid	C1=CC=C(C=C1)C(=O)O	HUC_49	Benzoic acid	OC(=O)c1ccccc1	Name
SPX_C32	Caffeine	CN1C=NC2=C1C(=O)N(C(=O)N2C)C	HUC_22	Caffeine	Cn1cnc2c1c(=O)n(C)c(=O)n2C	SMILES
SPX_C39	Codeine	CN1CC[C@]23C@?)4[C@H]1CC5=C2C(=C(C=C5)OC)O[C@H]3C@HO	HUC_153	Codeine	COc1ccc2c3c1OC1C43CCN(C)C2C4C=CC1O)C	Name
SPX_C23	Diethyl phthalate	CCOC(=O)C1=CC=CC=C1C(=O)OCC	HUC_24	Diethyl Phthalate	CCOC(=O)c1ccccc1C(=O)OCC	Name
SPX_C24	Diethylene glycol monobutyl ether	CCCCOCCOCCO	HUC_28	Diethylene glycol monobutyl ether	CCCCOCCOCCO	SMILES
SPX_C27	Dimethyl phthalate	COC(=O)C1=CC=CC=C1C(=O)OC	HUC_25	Dimethyl Phthalate	COC(=O)c1ccccc1C(=O)OC	Name
SPX_C40	Ethyl p-aminobenzoate	CCOC(=O)C1=CC=C(C=C1)N	HUC_159	Ethyl p-aminobenzoate	CCOC(=O)c1ccc(cc1)N	Name
SPX_C52	Fentanyl	CCC(=O)N(C1CCN(CC1)CCC2=CC=CC=C2)C3=CC=CC=C3	HUC_127	Fentanyl	CCC(=O)N(c1ccccc1)C1CCN(CC1)CCc1ccccc1	Name
SPX_C56	Hydrocortisone	C[C@]12CCC(=O)C=C1CCC3C@?)3(C@ ?)2C@HO	HUC_103	Hydrocortisone	OCC(=O)[C@@]1(O)CC(C@?)2[C@]1(C)CC@H[C@H]1[C@H]2CCC2=CC(=O)CC[C@]12C	Name
SPX_C57	Hydromorphone	CN1CC[C@]23(C@?)4[C@H]1CC5=C2C(=C(C=C5)O)O[C@H]3C(=O)CC4	HUC_154	Hydromorphone	O=C1CC(C@?)2[C@@]34[C@H]1Oc1c4c(C[C@H]2N(CC3)C)ccc1O	Name
SPX_C58	Ibuprofen	CC(C)CC1=CC=C(C=C1)C(C)C(=O)O	HUC_80	Ibuprofen	CC(Cc1ccc(cc1)C(C(=O)O)C)C	Name
SPX_C62	Ketoprofen	CC(C1=CC(=CC=C1)C(=O)C2=CC=CC=C2)C(=O)O	HUC_85	Ketoprofen	CC(C(=O)O)c1cccc(c1)C(=O)c1ccccc1	Name
SPX_C65	Methyl p-aminobenzoate	COC(=O)C1=CC=C(C=C1)N	HUC_160	Methyl p-aminobenzoate	COC(=O)c1ccc(cc1)N	Name

compound_id	compound_name	smiles	huskin_compound_id	huskin_compound_name	huskin_smiles	match_method
SPX_C68	Morphine	CN1CC[C@]23(C@?)4[C@H]1CC5=C2C(=C(C=C5)O)O[C@H]3C@HO	HUC_156	Morphine	O[C@H]1C=C(C@?)2[C@@]34[C@H]1Oc1c4c(C[C@H]2N(CC3)C)ccc1O	Name
SPX_C71	Nicotine	CN1CCC[C@H]1C2=CN=CC=C2	HUC_82	Nicotine	CN1CCCC1c1cccnc1	Name
SPX_C80	Progesterone	CC(=O)[C@H]1CC(C@?)2[C@@]1(CC[C@H]3[C@H]2CCC4=CC(=O)CC[C@]34C)C	HUC_111	Progesterone	O=C1CC[C@]2(C(=C1)CC(C@?)1(C@?)2CC[C@]2([C@H]1CC(C@?)2C(=O)C)C)C	Name
SPX_C83	Propylparaben	CCCOC(=O)C1=CC=C(C=C1)O	HUC_58	Propylparaben	CCCOC(=O)c1ccc(cc1)O	Name
SPX_C84	Resorcinol	C1=CC(=CC(=C1)O)O	HUC_96	Resorcinol	Oc1cccc(c1)O	Name
SPX_C87	Testosterone	C[C@]12CC[C@H]3C@HCCC4=CC(=O)CC[C@]34C	HUC_60	Testosterone	O=C1CC[C@]2(C(=C1)CC(C@?)1(C@?)2CC[C@]2([C@H]1CC(C@?)2O)C)C	Name
SPX_C90	Theophylline	CN1C2=C(C(=O)N(C1=O)C)NC=N2	HUC_89	Theophylline	Cn1c(=O)n(C)c2c(c1=O)[nH]cn2	Name
SPX_C93	Triclosan	C1=CC(=C(C=C1Cl)O)OC2=C(C=C(C=C2)Cl)Cl	HUC_235	Triclosan	Clc1ccc(c(c1)O)Oc1ccc(cc1Cl)Cl	Name
SPX_C102	p-Phenylenediamine	C1=CC(=CC=C1N)N	HUC_46	p-Phenylenediamine	Nc1ccc(cc1)N	Name

11.9 Supplementary Table S9. SkinPiX calibration by source

Table 19: SkinPiX cross-dataset calibration results by source and anchor-panel size.

reference_id	anchors	RMSE_uncalibrated	RMSE_calibrated	MAE_uncalibrated	MAE_calibrated	RMSE_improvement_percent	MAE_improvement_percent
SPX_R1	1	1.1585902	0.7146946	1.0205845	0.5624134	38.3134269	44.893010
SPX_R18	1	0.6629421	0.7077191	0.6180923	0.6315757	-6.7542855	-2.181453
SPX_R4	1	0.3932914	0.3519032	0.3831207	0.3001075	10.5235431	21.667642
SPX_R5	1	1.0251023	0.8950804	0.8794285	0.7599875	12.6838012	13.581661
SPX_R1	2	1.1585902	0.6570574	1.0205845	0.5065883	43.2882005	50.362923
SPX_R18	2	0.6629421	0.6767255	0.6180923	0.6420827	-2.0791141	-3.881357
SPX_R5	2	1.0251023	0.8850607	0.8794285	0.7502658	13.6612337	14.687117
SPX_R1	3	1.1585902	0.6231420	1.0205845	0.4749434	46.2155008	53.463589
SPX_R18	3	0.6629421	0.6575841	0.6180923	0.6376599	0.8082217	-3.165802
SPX_R5	3	1.0251023	0.8845855	0.8794285	0.7335328	13.7075916	16.589831
SPX_R1	4	1.1585902	0.6154457	1.0205845	0.4666579	46.8797761	54.275424
SPX_R5	4	1.0251023	0.8780141	0.8794285	0.7183699	14.3486427	18.314005
SPX_R1	5	1.1585902	0.5986220	1.0205845	0.4522462	48.3318610	55.687527

11.10 Supplementary Table S10. Wrong-source falsification by HuSkin reference

Table 20: Source-level falsification analysis comparing matched-source and different-source calibration.

reference_id	anchors	RMSE_uncalibrated	RMSE_matched_correct	RMSE_wrong	MAE_uncalibrated	MAE_matched_correct	MAE_wrong	RMSE_improvement_correct	RMSE_improvement_wrong
HUR_1	1	1.8602450	1.6284014	1.7193248	1.8602450	1.6284014	1.7193248	12.4630662	7.5753533
HUR_10	1	0.7165456	0.7017034	0.8626465	0.5884309	0.6041074	0.7473150	2.0713532	-20.3896104

reference_id	anchors	RMSE_uncalibrated	RMSE_calibrated	RMSE_correct	RMSE_wrong	MAE_uncalibrated	MAE_calibrated	MAE_correct	RMSE_improvement	MAE_improvement
HUR_11	1	1.4441848	0.9888120	1.4070804	1.4149938	0.9446015	1.3720007	31.5314774	2.5692255	31.5314774
HUR_13	1	1.3009987	1.3384597	1.4174321	1.2025977	1.2025977	1.2394369	-2.8794058	-8.9495406	-2.8794058
HUR_14	1	0.5096445	0.4298477	0.6253572	0.3891261	0.3758579	0.5540687	15.6573499	-22.7045933	15.6573499
HUR_15	1	0.8975273	0.8455532	1.0679844	0.8299301	0.7960059	0.9292456	5.7908120	-18.9918531	5.7908120
HUR_17	1	0.6188283	0.6921357	0.7697923	0.4022470	0.4730169	0.6274168	-11.8461558	-24.3951302	-11.8461558
HUR_18	1	1.9310926	2.1757050	2.0085425	1.9085000	1.9085000	1.9085000	-12.6670440	-4.0106754	-12.6670440
HUR_2	1	1.3835162	1.8681921	1.4134045	1.2113659	1.6816154	1.2449482	-35.0321763	-2.1603110	-35.0321763
HUR_20	1	0.6942592	0.8575528	0.8613210	0.6366629	0.7352525	0.7317581	-23.5205541	-24.0633282	-23.5205541
HUR_22	1	4.3089384	4.1202098	4.3298384	4.2783013	4.0881585	4.2988028	4.3799329	-0.4850391	4.3799329
HUR_24	1	0.6032846	0.7245196	0.8603331	0.6032846	0.7245196	0.8603331	-20.0958285	-42.6081616	-20.0958285
HUR_27	1	0.0213146	0.7465315	0.4927584	0.0213146	0.7465315	0.4927584	-3402.4484208	-2211.8392036	-3402.4484208
HUR_3	1	1.2162049	1.0753741	1.2957111	0.8510345	0.9638918	1.0129799	11.5795289	-6.5372439	11.5795289
HUR_31	1	1.0818841	1.2160437	1.2897711	0.8428719	0.9957111	1.0342884	-12.4005564	-19.2152809	-12.4005564
HUR_34	1	1.0615460	0.8217713	0.9489462	1.0615460	0.8217713	0.9489462	22.5873170	10.6071580	22.5873170
HUR_37	1	0.1098109	0.4219495	0.4677061	0.1098109	0.4219495	0.4677061	-284.2509406	-325.9194570	-284.2509406
HUR_39	1	1.0499628	0.6993047	0.9995640	0.9903067	0.6280085	0.9377590	33.3971938	4.8000572	33.3971938
HUR_40	1	0.4577840	0.0719866	0.5297447	0.4577840	0.0719866	0.5297447	84.2749907	-15.7193603	84.2749907
HUR_41	1	1.1532919	1.3263138	1.3796011	1.0779356	1.1073106	1.2296981	-15.0024362	-19.6228835	-15.0024362
HUR_48	1	0.6001922	0.7115582	0.7962547	0.4292958	0.6328911	0.6540829	-18.5550637	-32.6666257	-18.5550637
HUR_5	1	2.6411893	1.2342417	2.6237753	2.5490880	1.0076821	2.5271184	53.2694704	0.6593217	53.2694704
HUR_51	1	1.2431342	0.9385288	1.4392316	1.1848753	0.8450850	1.3794005	24.5030194	-15.7744372	24.5030194
HUR_52	1	0.6041333	0.6741869	0.7870015	0.5351009	0.5546020	0.6620127	-11.5957214	-30.2695211	-11.5957214
HUR_57	1	0.9169451	0.5745873	0.9605563	0.8122214	0.4865256	0.8811598	37.3367786	-4.7561437	37.3367786
HUR_62	1	1.3474212	0.7397185	1.3362964	1.3204707	0.6879236	1.3053633	45.1011700	0.8256353	45.1011700
HUR_78	1	2.8344732	2.4465879	2.9782507	2.8254689	2.4358807	2.9692796	13.6845649	-5.0724566	13.6845649
HUR_82	1	0.8478066	0.6682999	0.9734492	0.7013074	0.5656546	0.8783344	21.1730741	-14.8197180	21.1730741
HUR_83	1	2.1807769	1.8165759	2.3050905	2.1807769	1.8165759	2.3050905	16.7005179	-5.7004258	16.7005179
HUR_90	1	2.1170970	1.5094187	2.1815573	2.0752797	1.4501831	2.1378248	28.7033745	-3.0447476	28.7033745
HUR_94	1	0.5176389	0.6671715	0.5858888	0.4756489	0.6332724	0.5510033	-28.8874220	-13.1848380	-28.8874220
HUR_1	2	1.8602450	1.5331471	1.8112821	1.8602450	1.5331471	1.8112821	17.5835890	2.6320634	17.5835890
HUR_10	2	0.7165456	0.6785660	0.8774553	0.5884309	0.5862370	0.7619625	5.3003767	-22.4562996	5.3003767
HUR_11	2	1.4441848	0.8835119	1.2547071	1.4149938	0.8339116	1.2162389	38.8227925	13.1200507	38.8227925
HUR_14	2	0.5096445	0.4019126	0.6901178	0.3891261	0.3390713	0.6218726	21.1386386	-35.4116149	21.1386386
HUR_17	2	0.6188283	0.6793988	0.8176005	0.4022470	0.4466491	0.6852811	-9.7879361	-32.1207310	-9.7879361
HUR_18	2	1.9310926	2.2629902	2.0152136	1.9085000	1.9085000	1.9210007	-17.1870390	-4.3561359	-17.1870390
HUR_20	2	0.6942592	0.9138734	0.8784463	0.6366629	0.7651526	0.7551377	-31.6328917	-26.5300278	-31.6328917
HUR_24	2	0.6032846	0.7498089	0.8964233	0.6032846	0.7498089	0.8964233	-24.2877638	-48.5904500	-24.2877638
HUR_31	2	1.0818841	1.1196735	1.2407205	0.8428719	0.9544359	1.0045580	-3.4929290	-14.6814657	-3.4929290
HUR_34	2	1.0615460	0.7805857	1.0313376	1.0615460	0.7805857	1.0313376	26.4670899	2.8456998	26.4670899
HUR_37	2	0.1098109	0.5157370	0.5436736	0.1098109	0.5157370	0.5436736	-369.6590213	-395.0996949	-369.6590213
HUR_39	2	1.0499628	0.5974774	0.9831804	0.9903067	0.5329518	0.9245149	43.0953688	6.3604478	43.0953688
HUR_41	2	1.1532919	1.2015492	1.3734093	1.0779356	1.0920065	1.2146565	-4.1843041	-19.0860028	-4.1843041
HUR_48	2	0.6001922	0.8027389	0.9050359	0.4292958	0.7206758	0.7613146	-33.7469814	-50.7910105	-33.7469814
HUR_5	2	2.6411893	0.9859657	2.5875679	2.5490880	0.8639664	2.4894827	62.6696299	2.0301993	62.6696299
HUR_51	2	1.2431342	0.9116262	1.4079726	1.1848753	0.8209620	1.3442158	26.6671124	-13.2599057	26.6671124
HUR_52	2	0.6041333	0.6276984	0.9156278	0.5351009	0.5387657	0.7794803	-3.9006580	-51.5605782	-3.9006580
HUR_62	2	1.3474212	0.5919206	1.2807985	1.3204707	0.5327247	1.2493367	56.0701128	4.9444610	56.0701128
HUR_78	2	2.8344732	2.3749348	2.8794247	2.8254689	2.3641809	2.8702189	16.2124802	-1.5858853	16.2124802
HUR_82	2	0.8478066	0.6033061	1.0485110	0.7013074	0.5267674	0.9584756	28.8391832	-23.6733662	28.8391832

reference_id	n_anchors	RMSE_uncalibrated	RMSE_calibrated_correct	RMSE_calibrated_wrong	MAE_uncalibrated	MAE_calibrated_correct	MAE_calibrated_wrong	RMSE_improvement_correct	RMSE_improvement_wrong
HUR_83	2	2.1807769	1.6741154	2.4287497	2.1807769	1.6741154	2.4287497	23.2330769	-11.3708434
HUR_94	2	0.5176389	0.7089924	0.5908624	0.4756489	0.6740026	0.5578482	-36.9665982	-14.1456648
HUR_10	3	0.7165456	0.6567930	0.8880160	0.5884309	0.5684646	0.7663525	8.3389811	-23.9301467
HUR_11	3	1.4441848	0.8109849	1.4479310	1.4149938	0.7569725	1.4132775	43.8447947	-0.2594020
HUR_14	3	0.5096445	0.3671322	0.6979203	0.3891261	0.3323366	0.6236105	27.9630758	-36.9425875
HUR_17	3	0.6188283	0.6856633	0.9037716	0.4022470	0.4207497	0.7775966	-10.8002584	-46.0456188
HUR_20	3	0.6942592	0.9373739	0.8582217	0.6366629	0.7785480	0.7348190	-35.0178676	-23.6169098
HUR_24	3	0.6032846	0.7740621	0.7776233	0.6032846	0.7740621	0.7776233	-28.3079429	-28.8982541
HUR_31	3	1.0818841	1.0811769	1.3294021	0.8428719	0.9299684	1.0887221	0.0653595	-22.8784254
HUR_34	3	1.0615460	0.7743244	1.0639635	1.0615460	0.7743244	1.0639635	27.0569137	-0.2277323
HUR_37	3	0.1098109	0.5319507	0.4800079	0.1098109	0.5319507	0.4800079	-384.4241764	-337.1221801
HUR_39	3	1.0499628	0.5779113	1.0973499	0.9903067	0.5094986	1.0390086	44.9588801	-4.5132175
HUR_41	3	1.1532919	1.2083444	1.3613373	1.0779356	1.1079889	1.2161004	-4.7735057	-18.0392626
HUR_48	3	0.6001922	0.8220558	0.8405417	0.4292958	0.7352418	0.7101085	-36.9654257	-40.0454305
HUR_5	3	2.6411893	0.8995731	2.6168893	2.5490880	0.7703001	2.5196716	65.9406045	0.9200390
HUR_51	3	1.2431342	0.8545947	1.4405018	1.1848753	0.7630438	1.3766633	31.2548303	-15.8766109
HUR_52	3	0.6041333	0.5918326	0.8530157	0.5351009	0.5289499	0.7289568	2.0360873	-41.1966116
HUR_62	3	1.3474212	0.5144214	1.2976093	1.3204707	0.4609597	1.2636324	61.8217860	3.6968312
HUR_82	3	0.8478066	0.5618119	0.9784565	0.7013074	0.5007134	0.8950671	33.7334842	-15.4103365
HUR_83	3	2.1807769	1.6342725	2.3757428	2.1807769	1.6342725	2.3757428	25.0600780	-8.9402036
HUR_94	3	0.5176389	0.7414014	0.7185009	0.4756489	0.7127187	0.6840400	-43.2275275	-38.8034827
HUR_10	4	0.7165456	0.6476239	0.9244137	0.5884309	0.5625892	0.7969166	9.6186024	-29.0097469
HUR_11	4	1.4441848	0.7586197	1.3552818	1.4149938	0.7009467	1.3166993	47.4707338	6.1559246
HUR_14	4	0.5096445	0.3582150	0.7924465	0.3891261	0.3300470	0.7300653	29.7127618	-55.4900674
HUR_24	4	0.6032846	0.7773606	0.9561598	0.6032846	0.7773606	0.9561598	-28.8547035	-58.4923276
HUR_31	4	1.0818841	1.0409633	1.4185328	0.8428719	0.9358043	1.1715458	3.7823605	-31.1168965
HUR_34	4	1.0615460	0.7602826	1.0295102	1.0615460	0.7602826	1.0295102	28.3796906	3.0178478
HUR_39	4	1.0499628	0.5514992	1.1876384	0.9903067	0.4807370	1.1340542	47.4744078	-13.1124266
HUR_48	4	0.6001922	0.8544142	0.9017804	0.4292958	0.7595110	0.7560135	-42.3567649	-50.2486098
HUR_5	4	2.6411893	0.8241234	2.5491405	2.5490880	0.6913980	2.4487846	68.7972602	3.4851266
HUR_51	4	1.2431342	0.8243794	1.6295527	1.1848753	0.7335859	1.5709474	33.6854072	-31.0842184
HUR_62	4	1.3474212	0.4661661	1.3178972	1.3204707	0.4126540	1.2846107	65.4030888	2.1911507
HUR_82	4	0.8478066	0.5279409	1.2016658	0.7013074	0.4894183	1.1243583	37.7286209	-41.7381969
HUR_10	5	0.7165456	0.6343865	1.0289370	0.5884309	0.5566450	0.9007378	11.4659990	-43.5968592
HUR_11	5	1.4441848	0.7307444	1.1570872	1.4149938	0.6709878	1.1122326	49.4009042	19.8795595
HUR_14	5	0.5096445	0.3476294	0.8008516	0.3891261	0.3291176	0.7412729	31.7898146	-57.1392738
HUR_34	5	1.0615460	0.7666299	0.9488330	1.0615460	0.7666299	0.9488330	27.7817530	10.6178230
HUR_39	5	1.0499628	0.5199589	1.1480205	0.9903067	0.4488056	1.0949655	50.4783508	-9.3391642
HUR_48	5	0.6001922	0.8828268	1.1341355	0.4292958	0.7833066	0.9838509	-47.0906801	-88.9620626
HUR_5	5	2.6411893	0.7602199	2.4480074	2.5490880	0.6913980	2.3423579	71.2167589	7.3142008
HUR_62	5	1.3474212	0.4391080	1.2731534	1.3204707	0.3937167	1.2384408	67.4112332	5.5118468
HUR_82	5	0.8478066	0.5090093	1.3274114	0.7013074	0.4793415	1.2257371	39.9616330	-56.5700681

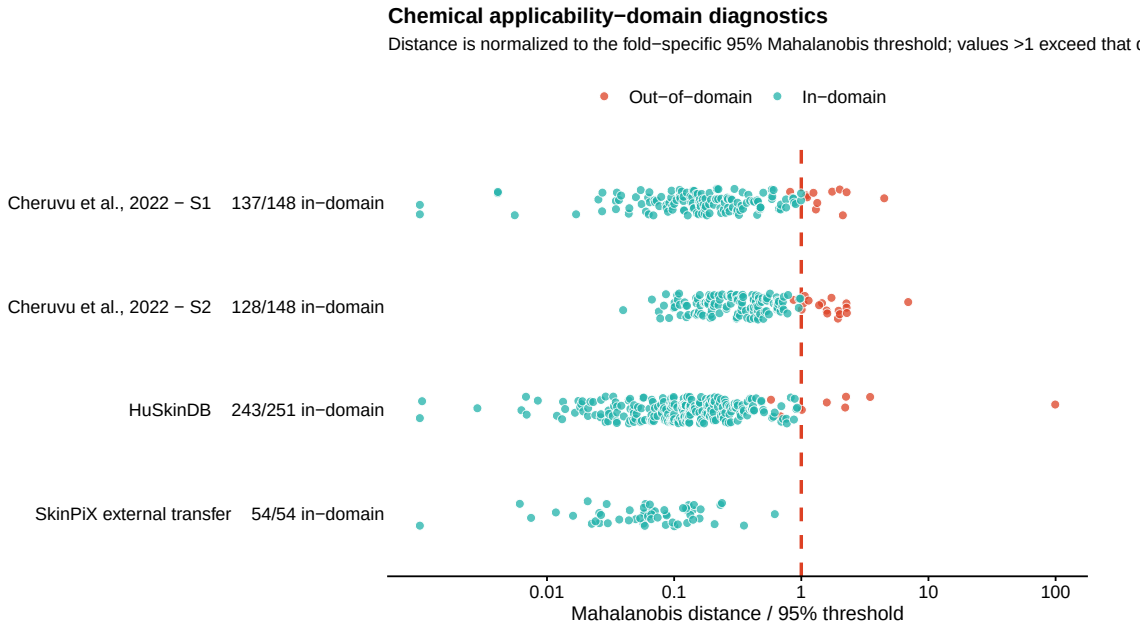


Figure 5: Applicability-domain diagnostics across Cheruvu S1, Cheruvu S2, HuSkinDB, and the external SkinPiX transfer set. Mahalanobis distance is normalized to the relevant 95% threshold; classification also requires all descriptors to remain within their training-set univariate ranges.

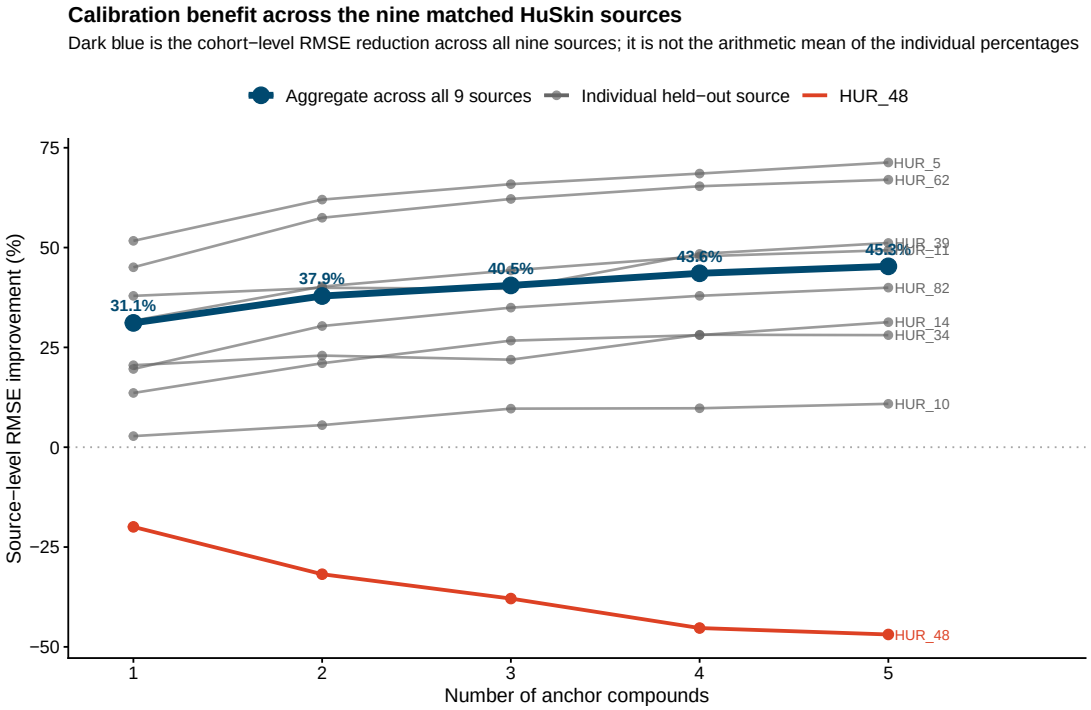


Figure 6: Calibration benefit across the matched HuSkin sources. Grey trajectories show individual held-out sources, the dark-blue trajectory is the cohort-level reduction in mean source-level RMSE, shading shows the 95% source-bootstrap interval, and HUR_48 is highlighted as the persistent calibration failure. The aggregate trajectory is not the arithmetic mean of the source-level percentage improvements.

11.11 Supplementary Figure S1. Applicability-domain diagnostics

11.12 Supplementary Figure S2. HuSkin source-level calibration heterogeneity

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